4.2 From relationships to access control

As discussed in chapter 2, digital identity is best understood not as a static profile or a set of credentials but as a relationship between two entities—for example, a student and a university, a contractor and a client organization, or a customer and a service provider. Identity systems exist to facilitate, mediate, and enforce the terms of these relationships. Authentication, as we saw in chapter 3, validates that relationship at a moment in time, proving that the entity interacting with the system is indeed one that's recognized in that relationship.

Authorization makes the relationship actionable and enforceable. It defines how a system determines what the authenticated entity can do, under what conditions, and with what data or resources. Access control is where identity becomes operational, turned into decisions that the system can enforce.

Here are a few examples that illustrate this transformation:

- Polly, a project manager, authenticates using her enterprise credentials. Her account includes a role that indicates she is a manager. The system authorizes her to view project dashboards and assign tasks—not simply because she authenticated, but because of her role within the organization.
- Carl, a call center agent, logs in with a passkey. A policy checks whether it's between 9 AM and 5 PM—his assigned shift. Even though he's authenticated, access to sensitive customer data is conditional on time. The system combines his identity with contextual data—his shift hours—to drive the authorization decision.
- Sally, a university student, signs in using SSO from her laptop. During enrollment, the MAC address of her device was recorded. A policy checks whether the request comes from a registered device. Access is granted not only based on her authentication but also because her session originates from a trusted device.

In each case, identity provides a contextual anchor; however, it's authorization that applies that context to determine whether access should be granted. The richness of the identity relationship—roles, attributes,